

ARJAN SINGH GILLAN

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EDUCATION

University of California, Irvine

B.S. in Biomedical Engineering | GPA: 3.61

Irvine, CA

Graduation: June 2028

EXPERIENCE

Applications & Clinical Engineering Intern

Irvine, CA

Eyedaptic — AR/AI Vision Assistance for Macular Degeneration

Jan 2026 – June 2026

- Ran testing and validation of an AI-driven vision-assistance platform on AR glasses, benchmarking 4 AR headsets and 4 AI agents on response quality and camera performance.
- Designed comparative evaluations of camera image quality across devices, documenting findings to guide product and engineering improvements.

Assistant Project Manager, University Rover Challenge

Irvine, CA

Legacy Robotics at UC Irvine

May 2025 – Present

- Oversee an 80-person engineering design team, spanning 5 subteams across multiple engineering majors, building a Mars rover for the 2027 University Rover Challenge.
- Manage a ~\$15K budget and ensure the rover meets the mechanical and electrical constraints of the competition across all subsystems.
- Developed an onboard soil biosignature assay to detect protein concentrations as a life indicator and authored the chemical safety and lab-handling protocol for onboard testing.

Undergraduate Research Assistant, Biomechanics

Irvine, CA

UCI Rehabilitation & Augmentation Lab

April 2025 – Present

- Conduct human gait rehabilitation trials applying spinal cord stimulation during walking and crisscross tasks, analyzing 3D motion-capture data in Vicon Nexus to extract joint kinematics and step metrics across baseline and stimulation sessions.
- Use MATLAB to quantify how movement direction and speed affect ankle proprioception, optimizing rehabilitation protocols for stroke-patient recovery.

Undergraduate Research Assistant, Neuroimaging

Irvine, CA

UCI Center for Neural Circuit Mapping — Xu Lab

Jan 2025 – Present

- Segment degu brain MRI scans in ITK-SNAP to build a novel neuroanatomical atlas and developed a program to quantify regional brain shrinkage for Alzheimer's research.

TECHNICAL PROJECTS

Bridgeless Eyewear System

Irvine, CA

Computer Aided Design and FEA Project

May 2026

- Co-designed a bridgeless eyewear system in SolidWorks that removes the nasal bridge from the load path, redistributing frame weight to the sides and back of the head through adjustable temple arms — for users recovering from nasal surgery or with saddle-nose and congenital nasal conditions.
- Validated a 58.9 g / 0.58 N 3D-printed prototype with static FEA (stress, strain, displacement) and hinge-fold and arm-extension motion studies, incorporating ball-detent sizing and a spring-loaded slide for adjustable, bridge-free fit.

Morse Code Learning Device — Capstone

Irvine, CA

Data Acquisition Project

Nov – Dec 2025

- Built a working Morse code learning device on Arduino in embedded C++: decoded button presses into dots/dashes with buzzer and LED feedback, mapped patterns to characters via a 48-entry lookup (letters, numbers, punctuation), and added two-player messaging plus a serial-to-Morse mode that plays typed messages back audibly.

Autocorrelogram Firing Program

Irvine, CA

Data Analysis Sandbox Project

Nov – Dec 2025

- Automated classification of neuronal firing patterns with a custom MATLAB tool, adding ACG visualization of electrophysiological datasets.

LEADERSHIP

Vice President of Finance

Irvine, CA

Engineering Student Council at UCI

March 2026 – Present

- Oversee a ~\$40K annual budget for a 22-member board, managing allocation and financial planning across council programs and events.

SKILLS & INTERESTS

Technical: Testing & Validation, Data Acquisition & Analysis, MATLAB, C++, C#, Arduino, ITK-SNAP, Vicon Nexus, Histology, PCB Design, Soldering, 3D Printing, CNC, CAD

Interests: Event Planning, Public Speaking, Cooking, Rock Climbing, Golf, Tennis